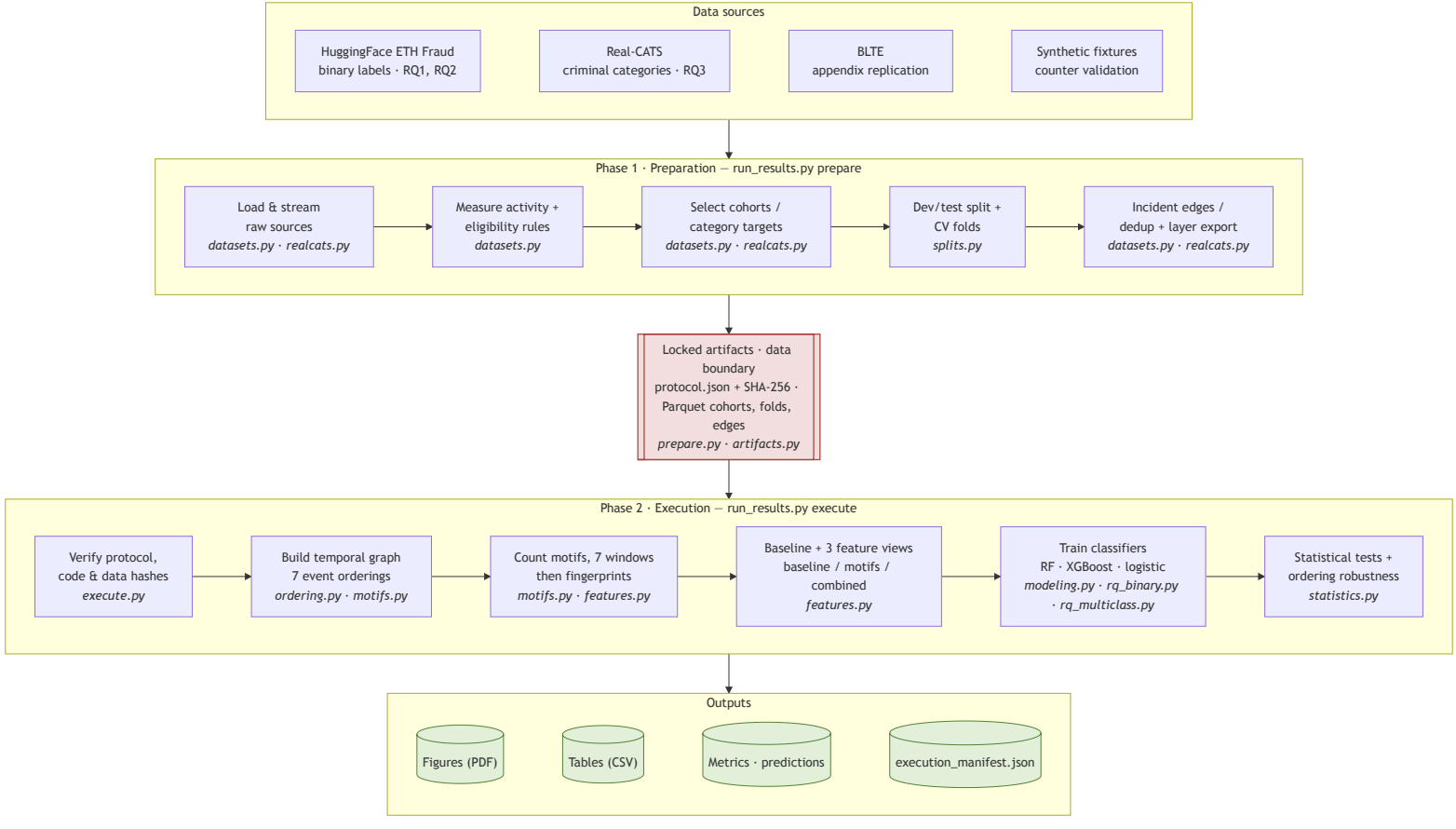
Data sources
HuggingFace ETH Fraud
binary labels · RQ1, RQ2
Real-CATS
criminal categories · RQ3
BLTE
appendix replication
Synthetic fixtures
counter validation
Phase 1 · Preparation — run_results.py prepare
Load & stream
raw sources
datasets.py · realcats.py
Measure activity +
eligibility rules
datasets.py
Select cohorts /
category targets
datasets.py · realcats.py
Dev/test split +
CV folds
splits.py
Incident edges /
dedup + layer export
datasets.py · realcats.py
Locked artifacts · data
boundary
protocol.json + SHA-256 ·
Parquet cohorts, folds,
edges
prepare.py · artifacts.py
Phase 2 · Execution — run_results.py execute
Verify protocol,
code & data hashes
execute.py
Build temporal graph
7 event orderings
ordering.py · motifs.py
Count motifs, 7 windows
then fingerprints
motifs.py · features.py
Baseline + 3 feature views
baseline / motifs /
combined
features.py
Train classifiers
RF · XGBoost · logistic
modeling.py · rq_binary.py
· rq_multiclass.py
Statistical tests +
ordering robustness
statistics.py
Outputs
Figures (PDF)
Tables (CSV)
Metrics · predictions
execution_manifest.json